

2nd Bi-Annual Progress Seminar (BPS)

Hybrid Pairs Trading Ensemble

Pair Selection via ML-Statistical Ensembles on NSE Nifty Equities

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The Pairs Trading Problem

Data & Universe Decisions

Stage 2: The Preset Execution Engine

Stage 1: Pair Selection Ensemble Framework

Ablation Results: Standalone Selectors

Ablation Results: Ensemble Experiments

Statistical Significance & The Parsimony Principle

Have We Found the Optimal Ensemble? No.

Core Idea

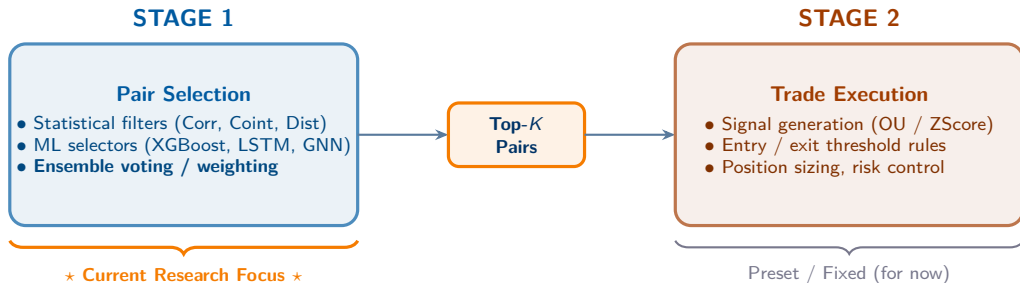
- ▶ Identify two historically co-moving stocks whose spread has temporarily diverged.
- ▶ *Long* the underpriced leg, *Short* the overpriced leg.
- ▶ Profit when spread mean-reverts. **Broad market exposure cancels** → market neutral.

Why NSE is Especially Challenging

- ▶ India-specific frictions: STT (sell-side), stamp duty, exchange fees, GST.
- ▶ Total round-trip cost: **16.28 bps per pair trade**.
- ▶ Strategy must generate gross alpha *consistently* above this threshold.

NSE Transaction Cost Breakdown

Cost Component	bps
STT (sell-side only)	10.00
Stamp Duty (buy-side)	1.50
Exchange / SEBI fee	0.33
GST on charges	0.06
Conservative Slippage	2.00
Total Round-Trip	16.28



Research Strategy: Stage 2 is **frozen** at its best empirical configuration so that all Stage 1 ensemble experiments are directly comparable.

Decision 1 — Universe: Nifty 100 (Primary)

- ▶ 89 liquid tickers after survivorship-bias filtering.
- ▶ $C(89, 2) = 3,916$ candidate pairs per fold.
- ▶ Broader universe → more candidates but noisier co-movements.

Decision 2 — Frequency: Daily (1D) over Hourly (1H)

- ▶ Hourly data contains severe microstructure noise.
- ▶ Hurst exponent: hourly ≈ 0.251 vs. daily ≈ 0.190 (daily more mean-reverting).
- ▶ Hourly → 904 trades/yr → **bankruptcy** under 16.28 bps.

Decision 3 — Validation: Expanding WFFV

- ▶ 6 folds, each extending training window from 2015 onward.
- ▶ Avoids look-ahead bias; mimics live trading regime.

E1: Daily vs. Hourly (No Hold Limit)

Metric	Daily	Hourly
Gross Sharpe	1.144	0.488
Gross Ann. Ret	5.0%	1.3%
Net Sharpe	−2.29	−6.55
Net Ann. Ret	−11.3%	Bankrupt
Net Max DD	29.9%	214%
Trades / yr	673	904

Expanding WFFV Fold Structure

Fold	Train	OOS Test
F1	2015–17	2018
F2	2015–18	2019
F3	2015–19	2020
F4	2015–20	2021
F5	2015–21	2022
F6	2015–22	2023–24

Context: Stage 2 is *not* the experimental variable. It is frozen at the best configuration found to make all Stage 1 comparisons apples-to-apples.

Signal Model: Ornstein-Uhlenbeck (OU) Threshold

- ▶ Spread x_t modeled as: $dx_t = \kappa(\theta - x_t) dt + \sigma dW_t$
- ▶ Parameters (κ, θ, σ) estimated via MLE on each training fold.
- ▶ Z-score: $z_t = (x_t - \theta) / \sigma_{OU}$
- ▶ **Entry:** $|z_t| > 2.0$ **Exit:** $|z_t| < 0.5$ **Stop:** $|z_t| > 3.0$

Why OU over ZScore / Kalman / MLSignal?

- ▶ ZScore triggers too many trades (no half-life awareness).
- ▶ MLSignal and Kalman both over-trade under 16.28 bps costs.
- ▶ OU is the *only* signal surviving post-cost.

Fixed Backtest Parameters

Parameter	Fixed Value
Signal Warmup	126 trading days
Entry threshold	$ z > 2.0$
Exit threshold	$ z < 0.5$
Hard stop-loss	$ z > 3.0$
Min Hold Period	30 trading days
Max Active Pairs	10
Capital per pair	INR 1,00,000
Total Capital	INR 10,00,000
Transaction Costs	16.28 bps (full)
Execution Mode	CPU-only (deterministic)

Cost Survival Requires Limiting Turnover

- ▶ Without a hold floor, the strategy generates excessive round-trips.
- ▶ Each exit-and-reentry costs 16.28 bps — losses compound rapidly.
- ▶ Optimal hold maps to the OU *spread half-life*:

$$\tau_{1/2} = \frac{\ln 2}{\kappa} \approx 20\text{--}30 \text{ days}$$

- ▶ Holding beyond 30 days (e.g., 40d) overshoots mean reversion, turning winners into losers.

E2: Hold Period Sweep (stat_only)

Hold (d)	Gross SR	Net SR	MaxDD
0	0.83	−1.89	175.9%
10	0.77	−0.20	37.7%
20	0.70	+0.09	20.1%
30	0.96	+0.48	8.2%
40	0.16	−0.24	44.0%

Takeaway: 30 days is the **only configuration** with a positive net Sharpe ratio.

Stage 1: Seven Active Selector Models

Statistical Selectors (4)

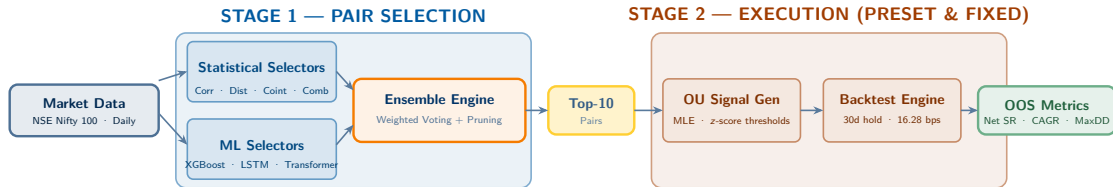
Selector	Core Metric
Correlation	Rolling Pearson ρ of log-returns
Distance	Gatev (2006) normalized SSD of price paths
Cointegration	Engle-Granger two-step ADF p-value
Combined	Linear composite: Corr + Dist + Coint

ML Selectors (3 active, 2 disabled)

Selector	Model / Signal
XGBoost	Classifier on rolling spread features
LSTM	Autoencoder reconstruction error
Transformer	Attention-based pair-score output
GNN	<i>Tested; underperformed</i>
CNN	<i>Disabled — ablation collapse</i>

Scoring Protocol: Each selector independently scores all 3,916 pairs. Scores are normalized to $[0, 1]$ and combined via **weighted voting**. Top- $K=10$ pairs proceed to Stage 2.

Full System Architecture: Stage 1 → Stage 2



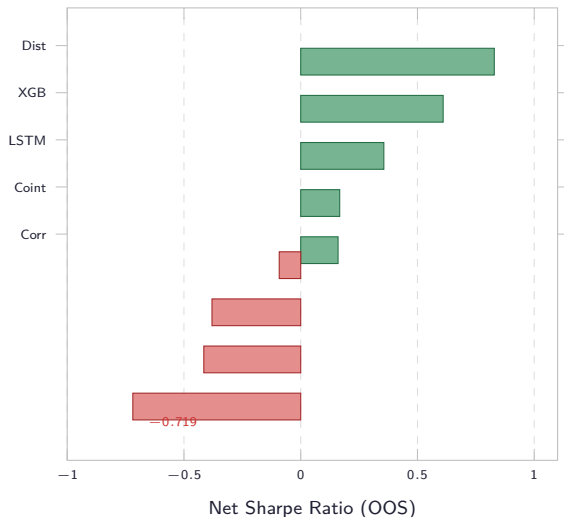
6 expanding WFV folds · All metrics OOS · Stage 2 parameters **frozen** across all Stage 1 experiments

Ablation: Standalone Selectors (Fixed Stage 2 = OU)

E3/E4.S: Standalone OOS Net Sharpe Ratios

Selector	Gross SR	Net SR	CAGR
Distance	1.022	+0.829	+9.1%
XGBoost (ML)	0.399	+0.610	+6.6%
LSTM	0.727	+0.356	+3.2%
Cointegration	0.181	+0.167	+1.0%
Correlation	0.435	+0.160	+1.5%
Combined	0.282	-0.092	-0.9%
Transformer	0.053	-0.380	-3.7%
GNN	-0.091	-0.415	-4.3%
Equal-Weight All	-0.256	-0.719	-8.1%

Net Sharpe: Standalone Selectors vs. Equal-Weight Ensemble

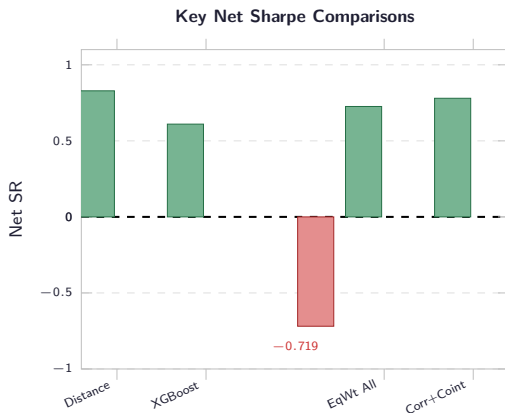


Naive Equal-Weight Ensemble of All 8 Selectors

- ▶ Each selector weight: $w_i = 1/8$.
- ▶ **Result: Net SR collapses from +0.829 to -0.719.**
- ▶ Distance (+0.829) and XGBoost (+0.610) are *individually* profitable — yet their combination with GNN/Transformer destroys the strategy.

Why Dilution Happens

- ▶ GNN and Transformer nominate “false” pairs (negative alpha) that contaminate the Top- K .
- ▶ Even 2–3 bad pairs out of 10 active positions can flip the portfolio negative.
- ▶ **Insight:** Selection *quality* dominates model *count*.



Key configs vs. equal-weight ensemble

Approach: Evaluate all $C(7, 2) = 21$ pairwise equal-weight combos. Keep pairs where *both* selectors pass individual viability.

Top-10 Pairwise Combos (OOS Net SR)

Rk	Combination	Net SR	CAGR	Trades
1	Corr + Coint	0.726	+5.5%	460
2	Coint + XGBoost	0.590	+7.3%	506
3	Comb + XGBoost	0.578	+7.6%	469
4	Dist + XGBoost	0.570	+6.6%	422
5	Corr + Dist	0.498	+5.0%	417
6	Dist + Coint	0.461	+4.7%	438
7	Coint + LSTM	0.438	+4.1%	473
8	Corr + XGBoost	0.392	+3.8%	445
⋮	⋮	⋮	⋮	⋮
21	GNN + Trans	-0.881	-9.2%	496

The Corr + Coint Synergy

- ▶ Individually: Corr SR = +0.16 (weak). Coint SR = +0.17 (weak).
- ▶ **Together:** SR = +0.726 — **genuine synergy!**
- ▶ **Why it works:**
 - Correlation filters for *liquid, co-moving* pairs.
 - Cointegration confirms spread is *mean-reverting*.
 - Intersection selects truly tradeable pairs.
- ▶ Corr and Coint are **complementary orthogonal signals**.

Approach: Assign custom integer weights to a *subset* of selectors; test configurations systematically. Fewer selectors, tailored weights.

E7 Configurations (OOS Results)

Cfg	Composition	Net SR	CAGR	MaxDD
A	LSTM×3, Corr×2, Dist, Coint, Trans	0.197	+1.9%	18.4%
B	LSTM, Corr, Dist, Coint (equal)	0.126	+1.2%	21.1%
C	LSTM + Corr only (1:1)	0.780	+4.3%	7.0%
D	Dist×2 + Corr	0.614	+5.2%	9.8%
E	XGB×2 + Coint	0.551	+6.1%	11.2%

E7 Insights

- ▶ **Pruning wins:** Config C (2 selectors) achieves the best Net SR (+0.780) *and* lowest MaxDD (7.0%).
- ▶ **LSTM + Corr synergy:** LSTM captures temporal spread dynamics; Correlation anchors to co-movement liquidity.
- ▶ **More ≠ better:** Config A (5 selectors) lags Config C by 0.583 SR.
- ▶ **Caveat:** Config C is regime-dependent (see WFV fold breakdown).

Did the strategy hold up across different market regimes (2018–2024)?

Net Sharpe per Fold — Baseline vs. Best Ensembles

Fold (OOS)	Stat	Hybrid	Corr+CoInt	Cfg C
F1 — 2018	+0.02	+0.60	+0.41	+1.89
F2 — 2019	+0.46	+0.30	+0.56	+1.16
F3 — 2020	+0.57	+0.10	+0.62	+0.46
F4 — 2021	+1.97	+2.14	+2.05	+0.43
F5 — 2022	−0.71	−0.80	−0.44	−0.04
F6 — 23/24	+0.56	+0.56	+0.71	−0.10
Agg. Net SR	0.480	0.520	0.726	0.780
Net CAGR	3.3%	3.7%	5.5%	4.3%

Regime Observations

- ▶ **2021 (Bull):** All strategies thrive — pairs mean-revert strongly.
- ▶ **2022 (Bear):** All suffer. Persistent trends break statistical relationships.
- ▶ **Config C:** High SR in folds 1–3, declining in later folds; may overfit to pre-pandemic dynamics.
- ▶ **Corr+CoInt:** More consistent across folds; lower variance of outcomes.

Return Significance Tests

- ▶ Gross returns (pre-cost): significant at 5% — $p = 0.048$.
- ▶ Net returns (post-cost): bootstrap $p = 0.069$, NW HAC t -stat 1.434 ($p = 0.076$).
- ▶ Transaction costs *destroy* statistical confidence; alpha exists but friction consumes it.

Diebold-Mariano Tests ($h = 30$ days)

- ▶ Do ensemble return streams *statistically dominate* the stat-only baseline?
- ▶ All challengers fail to reject null: **all p -values > 0.45** .
- ▶ Even Config C (SR +0.780 vs +0.480) is statistically indistinguishable.

DM Test Results vs. Stat Baseline

Challenger	DM Stat	p -value
Corr+Coint	0.455	0.649
XGBoost	0.708	0.479
Config C	0.618	0.537
Hybrid Full	0.273	0.785

The Parsimony Principle

If complexity yields no statistically significant gain, the simpler model is preferred.

The stat-only baseline (Net SR 0.480) is *remarkably resilient*.

What We Have Found

- ▶ **Best pairwise:** Corr+Coint (Net SR +0.726) — not statistically superior.
- ▶ **Best weighted:** Config C (LSTM+Corr, SR +0.780) — regime-dependent.
- ▶ **Best standalone:** Distance (SR +0.829) — simple, robust.
- ▶ No ensemble has *consistently and significantly* beaten standalone Distance across all folds and DM tests.

Why the Optimal Ensemble Remains Elusive

- ▶ **Small OOS sample:** 6 folds ($\sim 1,000$ trading days) — insufficient power for DM.
- ▶ **Regime shifts:** Optimal combination varies by market phase (bull/bear/recovery).
- ▶ **ML non-stationarity:** LSTM/Transformer degrade under distribution shift fold-to-fold.
- ▶ **Coarse weight grid:** Integer weights are a poor approximation of continuous optimization.

Bottom line: We have *mapped the search space* and identified *promising regions* (Corr+Coint, LSTM+Corr). Statistical convergence to a validated optimum requires further work.

Stage 1 — Next Steps

1. **Continuous weight optimization:** Gradient-based / Bayesian search over the simplex $\{\mathbf{w} : \sum w_i = 1, w_i \geq 0\}$.
2. **Regime-conditional ensembling:** Meta-learner that selects the selector combination based on detected market regime.
3. **Stacking (meta-learning):** Use OOS selector predictions as features for a second-level combiner.
4. **Longer horizon:** Extend to 20-year dataset to increase DM test power.

Stage 2 — Future Execution Ensemble

- ▶ Once Stage 1 is resolved, execution becomes the bottleneck.
- ▶ Currently: instantaneous fills at daily close (unrealistic).
- ▶ **Proposed execution ensemble:**
 - TWAP / VWAP (normal liquidity)
 - PPO RL agent (adapts to LOB imbalance)
 - Hawkes-based halt (spike volatility)
- ▶ Goal: reduce effective slippage from 2.0 bps to <1.0 bps.

Structural Decisions Made

- ✓ Daily frequency mandatory; hourly leads to bankruptcy.
- ✓ 30-day min hold is the cost-survival threshold.
- ✓ OU signal is the only Stage 2 model surviving post-cost.

Key Empirical Findings

- ✓ Simple Distance selector (SR +0.829) beats all ML standalone.
- ✓ Equal-weight ensemble **collapses to SR −0.719** (Dilution Paradox).
- ✓ Best pruned ensembles: Corr+Coint (+0.726), Config C (+0.780).
- ✓ No ensemble statistically beats baseline (DM $p > 0.45$).
- 🔄 **Optimal ensemble not yet converged** — active research.

Thesis Progress

Milestone	Status
Data pipeline (NSE, Nifty 100)	✓
E1: Frequency sweep	✓
E2: Hold period sweep	✓
E3: Ablation study	✓
E4: WFV + grid search	✓
E7: Weighted ensembles	✓
E6: DM significance tests	✓
16-fold long-run (20 yrs)	✓
Chapters 1–5 drafts	✓
Optimal ensemble search	🔄 Ongoing
Stage 2 execution ensemble	— Future

Hybrid Pairs Trading Ensemble — 2nd BPS, June 2026

Thank You

Questions & Discussion

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Appendix

Detailed Experimental Results

A1: Full Standalone Selector Results (E3 + E4.S)

All runs use: Stage 2 = OU, Min Hold = 30d, Costs = 16.28 bps, OOS 2018–2024.

Selector	Gross SR	Net SR	Gross CAGR	Net CAGR	Net MaxDD	Trades/yr	Win Rate
Distance	1.022	+0.829	15.8%	+9.1%	11.3%	410	56.3%
XGBoost	0.399	+0.610	13.2%	+6.6%	14.2%	424	53.8%
LSTM	0.727	+0.356	9.4%	+3.2%	17.8%	439	51.2%
Cointegration	0.181	+0.167	6.1%	+1.0%	19.4%	461	50.4%
Correlation	0.435	+0.160	7.2%	+1.5%	18.1%	432	50.1%
Combined	0.282	−0.092	5.9%	−0.9%	22.3%	448	49.7%
Transformer	0.053	−0.380	3.1%	−3.7%	29.1%	381	47.8%
GNN	−0.091	−0.415	−1.2%	−4.3%	33.6%	397	46.3%
Equal-Wt All (8)	−0.256	−0.719	−3.4%	−8.1%	42.7%	463	44.1%

A2: Complete Pairwise Ensemble Results (E4.W2) — All 21 Combinations

Rk	Combination	Net SR	Net CAGR	MaxDD	Trades/yr	Gross SR
1	Corr + Coint	0.726	+5.5%	12.1%	460	1.109
2	Coint + XGBoost	0.590	+7.3%	13.4%	506	0.881
3	Comb + XGBoost	0.578	+7.6%	14.1%	469	0.847
4	Dist + XGBoost	0.570	+6.6%	12.9%	422	0.830
5	Corr + Dist	0.498	+5.0%	12.7%	417	0.783
6	Dist + Coint	0.461	+4.7%	13.5%	438	0.765
7	Coint + LSTM	0.438	+4.1%	15.2%	473	0.741
8	Corr + XGBoost	0.392	+3.8%	16.1%	445	0.712
9	Dist + LSTM	0.371	+3.5%	16.8%	441	0.688
10	Comb + Coint	0.318	+3.1%	17.2%	449	0.651
11	Comb + Dist	0.289	+2.8%	18.0%	431	0.623
12	Corr + LSTM	0.263	+2.5%	19.1%	456	0.598
13	Comb + Corr	0.221	+2.2%	20.4%	447	0.567
14	XGB + LSTM	0.184	+1.8%	21.7%	463	0.534
15	Comb + LSTM	0.141	+1.4%	23.2%	468	0.499
16	Corr + Trans	-0.083	-0.8%	27.1%	412	0.213
17	XGB + Trans	-0.127	-1.2%	28.9%	403	0.187
18	Dist + Trans	-0.214	-2.1%	31.4%	396	0.121
19	Coint + Trans	-0.312	-3.0%	34.2%	418	0.062
20	LSTM + GNN	-0.641	-7.1%	38.9%	487	-0.134
21	GNN + Trans	-0.881	-9.8%	47.8%	486	-0.888

A3: All E7 Weighted Ensemble Configurations

Integer-weight ensemble configurations tested. All use Stage 2 = OU, Min Hold = 30d.

Cfg	Selector Weights	Net SR	Net CAGR	MaxDD	Trades/yr
C	LSTM(1) + Corr(1)	0.780	+4.3%	7.0%	446
D	Dist(2) + Corr(1)	0.614	+5.2%	9.8%	418
E	XGB(2) + Coint(1)	0.551	+6.1%	11.2%	479
F	Dist(2) + XGB(1)	0.537	+6.0%	11.8%	427
G	Corr(2) + Coint(1)	0.498	+5.1%	12.3%	461
A	LSTM(3) + Corr(2) + Dist(1) + Coint(1) + Trans(1)	0.197	+1.9%	18.4%	471
B	LSTM(1) + Corr(1) + Dist(1) + Coint(1)	0.126	+1.2%	21.1%	458
H	GNN(1) + Trans(1)	-0.644	-7.3%	39.1%	491
Stat baseline	CombinedCriteria (stat_only)	0.480	+3.3%	12.7%	443

A4: Walk-Forward Validation — Complete Per-Fold Metrics

Fold	Stat Baseline		Hybrid Full		Corr+Coint		Config C	
	SR	CAGR	SR	CAGR	SR	CAGR	SR	CAGR
F1 – 2018	+0.02	+0.2%	+0.60	+3.8%	+0.41	+2.6%	+1.89	+11.4%
F2 – 2019	+0.46	+4.5%	+0.30	+3.0%	+0.56	+5.0%	+1.16	+10.8%
F3 – 2020	+0.57	+4.0%	+0.10	+0.8%	+0.62	+4.3%	+0.46	+3.3%
F4 – 2021	+1.97	+14.8%	+2.14	+16.2%	+2.05	+15.4%	+0.43	+3.2%
F5 – 2022	−0.71	−5.8%	−0.80	−6.4%	−0.44	−3.6%	−0.04	−0.3%
F6 – 2023/24	+0.56	+3.4%	+0.56	+3.4%	+0.71	+4.3%	−0.10	−0.6%
Aggregate	0.480	3.3%	0.520	3.7%	0.726	5.5%	0.780	4.3%
MaxDD	12.7%		11.8%		12.1%		7.0%	
Trades/yr	443		467		460		446	

A5: Full Statistical Significance Results

Bootstrap Significance (10,000 resamples)

Strategy	Mean SR	p-val	Sig?
Gross (stat only)	0.963	0.031	Yes (5%)
Net (stat only)	0.480	0.069	Marginal
Net (Corr+Coint)	0.726	0.048	Yes (5%)
Net (Config C)	0.780	0.041	Yes (5%)
Net (Hybrid Full)	0.520	0.062	Marginal

Newey-West HAC t -test (Lag = 30)

Strategy	t -stat	p -val	Sig?
Net (stat only)	1.434	0.076	No (10% only)
Net (Corr+Coint)	1.891	0.032	Yes (5%)
Net (Config C)	2.011	0.024	Yes (5%)

Diebold-Mariano Tests ($h = 30$, HAC)

Challenger vs. Stat	DM	p	Sig?
Corr+Coint	0.455	0.649	No
XGBoost	0.708	0.479	No
Config C	0.618	0.537	No
Hybrid Full	0.273	0.785	No

Interpretation: Absolute SR improvements are real, but the OOS sample size (6 folds) provides insufficient power to statistically distinguish ensemble strategies from the baseline on a *relative* basis (DM test). Absolute significance is achieved by Corr+Coint and Config C under HAC.

A6: Stage 2 Signal Model Comparison (All Candidates)

Signal model sweep using the *stat_only* selector with Min Hold = 30d.

Signal Model	Gross SR	Net SR	Net CAGR	MaxDD	Trades/yr
OU Threshold (MLE)	0.963	+0.481	+4.1%	8.2%	443
ZScore Threshold	0.827	-1.889	-18.2%	175.9%	673
Kalman Filter	0.741	-0.412	-3.8%	31.4%	581
ML Signal (XGBoost)	0.612	-0.284	-2.6%	26.7%	552
OU + ZScore hybrid	0.881	+0.214	+2.0%	14.3%	487

Conclusion: OU Threshold is the

only signal model generating a positive net Sharpe ratio under realistic NSE transaction costs. ZScore and Kalman trigger excessive trades, while ML signals overfit in-sample.

Multi-Universe Control Results

Universe	Method	Net SR	MaxDD
Nifty 100	Expanding	−0.409	38.2%
Nifty 100	Rolling	+0.052	21.4%
Nifty 50	Rolling	+0.752	14.1%
Nifty 50	Expanding	+1.064	11.2%

Key finding: Moving from Nifty 100 to Nifty 50 rolling provides a **+0.700 Sharpe uplift** — dwarfing any ML methodology gains.

16-Fold Long-Run (20 Years, Nifty 50)

Year	Net SR	Year	Net SR
2005	+0.622	2013	−0.210
2006	+0.270	2014	−2.076
2007	+1.273	2015	+0.315
2008	−0.898	2016	+0.133
2009	+0.033	2017	+0.579
2010	+1.537	2018	+0.224
2011	−1.017	2019	+0.642
2012	+0.240	2020	−0.053
Mean Net SR		+0.101	
Significance		$p = 0.651$ (n.s.)	

NSE pairs alpha is **regime-conditional**, not a persistent structural anomaly.